Can AI Save Lives?

Building an Intelligent Clinical Early Warning System

Deep Learning — Complex Computing Problem

Department of Artificial Intelligence

Dataset: Patient Survival Prediction (Kaggle) — 64,199 ICU patients

Report — Part B

Section 1: The Clinical Problem

The Challenge of Predicting Patient Deterioration

Predicting patient deterioration is categorically harder than standard classification tasks, and understanding why is crucial before any model is trusted with a life. In a typical supervised learning problem, spam detection, image labelling, fraud identification, the input data is relatively static, the classes are well-separated, and a wrong prediction carries limited consequence. In an ICU, none of these conditions hold. A patient's condition evolves continuously, the boundary between stable and deteriorating is physiologically blurred and time-dependent, and a single missed prediction can result in death.

The dataset used in this project, the Patient Survival Prediction dataset from Kaggle, comprising 64,199 ICU admissions, reflects this difficulty directly. Only 8.6% of patients died (5,540 out of 64,199), creating a severe class imbalance. A naive classifier that always predicts survival would achieve 91.4% accuracy while being clinically useless, having identified zero patients at risk. This illustrates why accuracy is a misleading metric in this domain and why the problem demands both technical rigour and domain awareness.

Beyond class imbalance, the nature of deterioration itself presents a modelling challenge. Vital signs fluctuate naturally. A temporary rise in heart rate may reflect anxiety, pain, or activity, not sepsis. The signal exists in the *pattern of change over time* and in the *combination* of multiple indicators, not in any single measurement. This is precisely what distinguishes an experienced ICU clinician from a junior observer: the ability to synthesise multiple evolving signals into a coherent picture of risk.

Why Vital Signs Alone Are Insufficient

Tabular vital-signs data captures what can be measured numerically: heart rate, blood pressure, respiratory rate, temperature, SpO2. These are invaluable but fundamentally incomplete. A heart rate of 120 bpm means something very different in a 24-year-old post-operative patient than in a 75-year-old with a history of sepsis and hepatic failure. The number is identical; the clinical meaning is not.

Unstructured clinical text, physician notes, nursing observations, triage summaries, provides the contextual layer that numbers cannot. A note may describe a patient as *"increasingly confused and diaphoretic despite stable vitals"*, a description that contains no measurable number yet communicates immediate danger to any trained clinician. It captures subtle observations: the quality of breathing, the patient's affect, the physician's gestalt impression of deterioration. It records comorbidity context, medication changes, and diagnostic reasoning that never appear in a vital-signs spreadsheet.

In this project, since the dataset does not contain free-text notes, we constructed structured clinical narratives from each patient's available features, demographics, diagnosis systems, APACHE scores, comorbidities, and multi-day vital sign trajectories. This approach, standard in clinical NLP research, allows us to demonstrate the full Transformer-based pipeline and validate that a language model can extract meaningful clinical signal from prose descriptions of patient state.

Why Recall Matters More Than Accuracy

In binary classification, Recall (also called sensitivity) measures the proportion of actual positive cases, deteriorating patients, that the model correctly identifies. It is defined as TP / (TP + FN), where FN represents patients who were deteriorating but whom the model classified as stable. Precision measures the proportion of positive predictions that were correct. Accuracy measures overall correctness across both classes.

In a healthcare early warning system, the asymmetry of error costs makes Recall the primary clinical metric. Consider a concrete scenario: a 67-year-old patient, Mrs. A, is admitted to the Med-Surg ICU following emergency abdominal surgery. Over the first twelve hours, her heart rate climbs incrementally, her MAP drops gradually, and her SpO2 begins to trend downward, individually borderline, collectively alarming. The early warning model processes her latest vitals.

If the model produces a **false negative**, predicting she is stable when she is not, no alert is raised, no physician is called, and Mrs. A progresses to septic shock unobserved. The intervention window closes. She may die. The cost of this error is irreversible.

If the model produces a **false positive**, flagging her as at-risk when she is stable, a physician is called unnecessarily. The physician assesses her, finds her stable, and returns to their duties. The cost of this error is a few minutes of a clinician's time and mild disruption. This is recoverable.

This asymmetry, irreversible harm from false negatives versus recoverable inconvenience from false positives, is why every model in this project was evaluated with Recall as the primary clinical metric. An early warning system that achieves 95% accuracy but misses 60% of deteriorating patients is not a safety tool; it is a liability. The goal is to catch as many true deteriorations as possible, even at the cost of some additional false alarms.

Section 2: Baseline Design Decisions

The Vanishing Gradient Problem

Deep neural networks are trained through backpropagation: gradients of the loss function are computed with respect to each parameter via the chain rule and propagated backward through the network. In a deep network, this requires multiplying gradients across each layer. When activation functions like sigmoid or tanh are used, their derivatives are bounded between 0 and 0.25. Multiplying these small values across many layers causes gradients to shrink exponentially, approaching zero before reaching the early layers. The result is that the first layers of a deep network learn almost nothing, and the network fails to capture hierarchical feature representations.

In this project, all hidden layers use the **ReLU activation function** (Rectified Linear Unit: $f(x) = \max(0, x)$). ReLU has a gradient of exactly 1 for all positive inputs, which means gradients propagate backward through ReLU layers without attenuation. This simple property makes deep networks trainable. The **He normal initialisation** was used for all weight matrices, which is specifically designed to preserve the variance of activations when ReLU is the activation function, further stabilising the forward and backward passes.

Batch Normalisation, applied after each hidden layer, addresses a complementary problem known as internal covariate shift, the tendency for the distribution of each layer's activations to shift during training as the parameters of earlier layers change. By normalising activations to zero mean and unit variance per mini-batch (then applying learned scale and shift parameters), BatchNorm stabilises training, permits higher learning rates, and provides mild regularisation. In our experiments, removing BatchNorm increased validation loss and reduced Recall from 0.5681 to 0.6499 in the SGD variant, though with lower precision, demonstrating its role in stabilising the optimisation landscape.

Optimiser Comparison: Adam vs SGD

Two optimisers were trained under identical architectural and regularisation conditions. **Adam (Adaptive Moment Estimation)** maintains per-parameter adaptive learning rates by tracking the first moment (mean of gradients) and second moment (uncentred variance). This allows it to take larger steps for parameters with historically small gradients and smaller steps for frequently updated parameters. With a learning rate of 0.001, Adam converged faster and achieved the highest Recall of 0.5681 with Accuracy 0.8618 and F1 of 0.4189.

SGD with momentum (learning rate 0.01, momentum 0.9) accumulated gradient history across steps, helping to escape shallow local minima and accelerate movement in consistent gradient directions. It required more epochs to converge and produced lower Accuracy (0.8301) but achieved a Recall of 0.6583, higher than Adam, suggesting it found a region of parameter space more sensitive to the minority class. This is consistent with the known tendency of SGD to find flatter minima that generalise differently from Adam's sharper but often better-performing solutions.

For clinical deployment, Adam was selected as the preferred baseline optimiser based on its superior F1-Score (0.4189 vs 0.4046) and more stable convergence, while noting that the recall trade-off between the two is clinically significant.

Loss Function vs Cost Function

These terms are related but distinct. The **loss function** is defined per sample: it quantifies the error of a single prediction against its true label. In this project, binary cross-entropy is used: $L = -(y \log(p) + (1-y) \log(1-p))$, where y is the true label and p is the predicted probability. The **cost function** is the aggregate of the loss function over the entire training set, typically the mean loss across all samples in a batch or epoch. It is the cost function that the optimiser minimises, not the per-sample loss. Batch Normalisation and Dropout affect the cost function landscape, making it smoother and more generalisable, which is why they both improve convergence and reduce overfitting.

Dropout as a Regularisation Strategy

Dropout randomly sets a fraction of neuron activations to zero during each training step (rate = 0.3 in this project). This prevents any single neuron from becoming indispensable, the network is forced to learn distributed, redundant representations across multiple pathways. During inference, all neurons are active but their weights are scaled down by the dropout rate, providing an implicit ensemble effect.

The ablation study confirmed its value: the DNN trained without Dropout showed marginally higher Accuracy (0.8463 vs 0.8618 for Adam) but lower Recall (0.5849 vs 0.5681, a relatively small difference in this direction), and the gap was more visible in the validation loss curves, which showed faster divergence without Dropout, indicating overfitting. In a healthcare context where the model will be deployed on patients from populations potentially different from the training distribution, overfitting is a patient safety concern, not merely a statistical one.

Generation 1 Results Summary

Model	Accuracy	Precision	Recall ★	F1-Score
DNN (Adam)	0.8618	0.3317	0.5681	0.4189
DNN (SGD)	0.8301	0.2921	0.6583	0.4046
DNN (No Dropout)	0.8463	0.3043	0.5849	0.4003
DNN (No BatchNorm)	0.8292	0.2892	0.6499	0.4003

Section 3: Modelling Patient Timelines

Vanishing Gradients in Recurrent Networks

In a standard RNN, the hidden state at each timestep is computed as $h_t = \tanh(W_h * h_{t-1} + W_x * x_t)$. Backpropagation through time requires multiplying gradients across every timestep in the sequence. Since the tanh derivative is bounded between 0 and 1, and is multiplied by the weight matrix W_h repeatedly, the gradient with respect to parameters at early timesteps decays exponentially. For sequences of even moderate length (10-20 steps), gradients effectively vanish, making it impossible for the RNN to learn dependencies between distant timesteps, a critical failure when patient deterioration often develops over hours or days.

LSTM Gates and the Cell State Solution

The Long Short-Term Memory architecture addresses vanishing gradients through a dedicated **cell state C_t** that runs linearly through the sequence with only minor interactions, and three learned gates that control information flow. The **forget gate** ($f_t = \mathrm{sigmoid}(W_f * [h_{t-1}, x_t])$) decides what fraction of the previous cell state to retain. The **input gate** (i_t) decides what new information to write into the cell state. The **output gate** (o_t) determines what portion of the cell state to expose as the hidden state h_t. Because the cell state update involves addition rather than repeated multiplication, gradients can flow backward through time without exponential decay — allowing the LSTM to maintain clinically relevant information across extended patient timelines.

The GRU (Gated Recurrent Unit) simplifies this architecture by merging the forget and input gates into a single **update gate** and eliminating the separate cell state. This reduces the parameter count by approximately 25% compared to an equivalent LSTM. In our experiments, the GRU achieved a Recall of 0.7799 in 74.7 seconds, marginally higher Recall than the LSTM (0.7652 in 59.8 seconds) while requiring more training time, an apparently paradoxical result explained by the difference in gradient landscapes during training. For a clinical monitoring system where inference speed matters, the GRU's reduced complexity offers a practical advantage without meaningful sacrifice in predictive performance.

Unidirectional vs Bidirectional LSTM for Real-Time Monitoring

The Bidirectional LSTM achieved the highest Accuracy (0.7820) and a competitive F1-Score (0.3784) in offline evaluation. It does so by processing each sequence in both forward (t=0 to t=T) and backward (t=T to t=0) directions, concatenating the resulting hidden states to give the classification head access to both past and future context. This is architecturally powerful for retrospective analysis.

However, it is fundamentally **inappropriate for real-time ICU deployment**. In a live early warning system, future timesteps, the vital signs that will be recorded in the next hour or the next day, do not yet exist. A Bidirectional LSTM cannot produce a prediction at time t without having seen t+1, t+2, and all subsequent measurements. Deploying it in a live system would require waiting until the patient's full admission trajectory is complete before issuing any alert, at which point the intervention window may have long closed.

The unidirectional LSTM, by contrast, can issue a prediction after each new measurement arrives, t=0, then t=1, then t=2, making it the only architecturally appropriate choice for real-time patient monitoring. The Bidirectional LSTM remains valuable for post-discharge risk audits, retrospective deterioration analysis, and model research, but its offline accuracy advantage is irrelevant if the model cannot be deployed in the environment where lives are at stake.

Generation 2 Results Summary

Model	Accuracy	Precision	Recall ★	F1-Score	Train Time
LSTM	0.7721	0.2445	0.7652	0.3706	59.8s
GRU	0.7735	0.2482	0.7799	0.3765	74.7s
Bi-LSTM	0.7820	0.2523	0.7568	0.3784	97.2s

Section 4: The Transformer and Clinical Language

Transfer Learning and ClinicalBERT

Training a language model from scratch on a small clinical dataset is impractical: capturing the statistical structure of medical language requires exposure to millions of documents. ClinicalBERT (*emilyalsentzer/Bio_ClinicalBERT*) was pre-trained on over two million clinical notes from the MIMIC-III database using a masked language modelling objective, predicting randomly masked tokens from context. This process instills deep knowledge of medical vocabulary, clinical abbreviations, physiological relationships, and the semantic patterns that distinguish stable from deteriorating patient descriptions.

Transfer learning in this context means that ClinicalBERT arrives at our task already knowing, for instance, that "SpO2 of 77%" is more alarming than "SpO2 of 97%", that "sepsis" and "hypotension" frequently co-occur in high-acuity contexts, and that GCS scores below 8 indicate profound neurological compromise. Training from scratch on 8,000 patient notes, our fine-tuning subset, would produce a model with no such prior knowledge, and would require orders of magnitude more data to approach equivalent performance. Transfer learning collapses this data requirement by borrowing representations that took weeks of pre-training to build.

The Self-Attention Advantage

Traditional sequence models, LSTMs included, process text token by token, left to right. This means that when the model reaches the word "sepsis" in a long clinical note, its representation of earlier tokens has been compressed into a fixed-size hidden state. Relationships between distant tokens are difficult to capture.

BERT's self-attention mechanism computes a weighted relationship between every token and every other token in the sequence simultaneously. For each token, attention weights determine how much it should "attend to" every other position. This is analogous to how a consultant physician reads a clinical note: they do not parse it word by word but scan directly for the highest-signal terms, "intubated", "hypotension", "MAP 54", "Sepsis", and weight their interpretation accordingly. Self-attention gives the model this same non-sequential, relevance-driven reading capability, which is architecturally well-suited to the way clinical meaning is distributed across notes.

What the Attention Heatmaps Revealed

Attention weights were extracted from the final transformer layer, averaged across all 12 attention heads, and the [CLS] token row was taken as the model's summary attention distribution, representing what the model attended to when forming its classification decision.

The **frozen ClinicalBERT** model attended most strongly to structural tokens: [SEP] (highest weight, ~0.10), [CLS], followed by clinically meaningful terms including "monitoring", "oxygen", "eyes", "scale" (fragments of Glasgow Coma Scale), "signs", "verbal", "sat" (oxygen saturation), and "admission". The presence of GCS-related tokens ("eyes", "scale", "verbal") and oxygenation terms ("oxygen", "sat") in the top positions is clinically coherent, these are precisely the indicators clinicians prioritise when assessing acute deterioration risk. The frozen model is using ClinicalBERT's pre-trained knowledge without adaptation, and that knowledge directs it toward the right signals.

The **full fine-tuned** model shows a different pattern: the highest attention weight (~0.31) was on "recovery", followed by "room" (~0.12) and [SEP] (~0.10). The phrase "Operating Room / Recovery" appears in the icu_admit_source field for this patient, suggesting the model learned to weight admission source heavily — a clinically valid feature, as direct post-operative admissions carry different risk profiles than emergency admissions. The numeric tokens (104, 78, 3) likely correspond to Apache diagnosis codes and vital sign values, indicating the full fine-tuned model adapted its attention to task-specific numeric clinical identifiers.

Both patterns suggest the models are attending to clinically meaningful features rather than spurious correlations, which is a necessary (though not sufficient) condition for safe clinical deployment. The interpretability advantage of attention, the ability to show a clinician

"the model flagged this patient because of falling SpO2 and low GCS verbal score", is a critical requirement for clinician trust that DNN and LSTM models cannot provide.

Frozen vs Full Fine-tuning

The empirical comparison revealed a striking trade-off. **Frozen ClinicalBERT** achieved high Accuracy (0.9017) but very low Recall (0.0952) and F1 (0.1449) in 279.2 seconds. It is essentially classifying most patients as survived, leveraging the class imbalance to score well on accuracy while failing completely on the clinical objective. The frozen base cannot adapt its representations to our specific task, it applies MIMIC-III-derived knowledge as-is, and while that knowledge is relevant, without fine-tuning the classification head alone cannot redirect the model's attention toward mortality-specific signals.

Full fine-tuning produced dramatically better clinical performance: Recall of 0.6857 and F1 of 0.2903, at the cost of 645.8 seconds of training (2.3x longer) and lower Accuracy (0.7067). All 110 million parameters were updated, allowing the model to reshape its internal representations around the mortality prediction objective. The lower accuracy reflects a more aggressive positive prediction strategy, the model correctly catches more deteriorating patients at the expense of more false alarms, which is precisely the clinical trade-off we want.

Frozen fine-tuning is preferable when the dataset is very small (under 2,000 labelled samples), the computational budget is severely constrained, or the downstream task is closely aligned with the pre-training domain such that minimal adaptation is needed. Full fine-tuning justifies its cost when sufficient labelled data is available (as in our 8,000-sample subset), a GPU is accessible, and maximum clinical sensitivity is the objective, which is always the case in a life-critical early warning system.

Positional Encoding

Unlike LSTMs, which process tokens sequentially and therefore have implicit positional awareness, the Transformer processes all tokens in parallel. Without explicit positional information, the model would treat "heart rate elevated, blood pressure normal" identically to "blood pressure elevated, heart rate normal", losing the sequential structure of the note. Positional encodings are added to each token's embedding before the attention layers, injecting information about where each token appears in the sequence. For clinical notes, this matters because the order of clinical observations carries meaning: findings described early in a note (chief complaint, admission vitals) often carry different weight than observations made later (day-one monitoring, discharge planning notes).

Parallelisation and Hospital-Scale Deployment

One of the most operationally important properties of the Transformer architecture is that self-attention computations across all token positions are performed in parallel, unlike LSTMs, which process sequences step by step and cannot be parallelised across the time dimension. On modern GPU hardware, this means a single ClinicalBERT forward pass for a 128-token clinical note takes milliseconds.

For a hospital system processing thousands of patient notes per hour, this parallelism is not a theoretical advantage, it is a deployment requirement. An LSTM-based system processing 10,000 notes sequentially would introduce latency that could delay alerts by minutes. With BERT running on a multi-GPU inference cluster, the same 10,000 notes can be processed in parallel batches, with results returned in near real-time. This scalability makes Transformer-based models the architecturally appropriate choice for hospital-grade deployment, provided the inference infrastructure is available.

Generation 3 Results Summary

Model	Accuracy	Precision	Recall ★	F1-Score	Train Time
ClinicalBERT (Frozen)	0.9017	0.3030	0.0952	0.1449	279.2s
ClinicalBERT (Full)	0.7067	0.1841	0.6857	0.2903	645.8s

Section 5: Deployment Verdict

Recommended Model for ICU Deployment

Across six models evaluated in this project, the deployment recommendation is **GRU** as the primary real-time alerting model, with **ClinicalBERT (Full Fine-tune)** as a complementary offline risk stratification layer.

The GRU achieved the highest Recall of any deployable model at 0.7799, training in 74.7 seconds with a compact architecture that supports low-latency inference. It captures temporal dynamics in patient vitals,a fundamentally more appropriate inductive bias than the DNN's static snapshot, and does so more efficiently than the LSTM (0.7652 Recall, 59.8s) with a comparable parameter count. Its unidirectional processing makes it compatible with real-time data streams, and its interpretable hidden states can be inspected to understand which timestep drove a particular alert.

The DNN (Adam) achieved the best F1-Score in Generation 1 (0.4189) and is appropriate as a fast triage tool where only a single snapshot of vitals is available. The ClinicalBERT Full model's Recall of 0.6857 is competitive, but its 645.8-second training time and higher inference cost make it unsuitable as the primary real-time alerting engine. It is better positioned as a *secondary review layer*, processing narrative summaries of patient state for scheduled risk stratification rounds, where latency is acceptable and the added explainability of attention weights provides clinical value.

The Bidirectional LSTM, despite its offline accuracy advantage, is excluded from real-time deployment for the architectural reasons discussed in Section 3. The frozen ClinicalBERT model, with a Recall of 0.0952, is clinically unacceptable and should not be deployed in any patient-facing capacity.

Ethical Concerns in Clinical AI Deployment

Three categories of ethical concern demand attention before any model from this project reaches a clinical environment.

Bias and health equity. The training dataset contains demographic features including ethnicity and gender. If the training population is not representative of the deployment population, for instance, if certain ethnic groups are underrepresented in the ICU cohort, the model may have systematically lower Recall for those groups, meaning deteriorating patients from minority populations are more likely to be missed. This is not merely a technical failure; it is a health equity failur. Deployment requires bias auditing across demographic subgroups, and retraining on locally representative data where disparities are identified.

Privacy and data governance. Clinical notes contain highly sensitive personal health information. Using patient data to train and fine-tune models requires explicit governance frameworks: informed consent or institutional review board approval, data anonymisation protocols, secure computation environments, and strict access controls. The pseudo-note generation approach used in this project mitigates some direct privacy risk, but any deployment using real clinical text must comply with applicable health data regulations.

Accountability and the human-in-the-loop requirement. An AI early warning system should augment clinical judgement, not replace it. No model in this project should autonomously trigger clinical interventions. Every alert must be reviewed by a qualified clinician who retains decision authority. Clear accountability chains must be established for cases where the model fails, either missing a deterioration or generating excessive false alarms that cause alert fatigue. The system must be continuously monitored in deployment, with performance metrics tracked at the patient population level to detect distributional shift as patient demographics and care practices evolve.

Extending to Chest X-Ray Analysis: A Multimodal Architecture

If the hospital wished to incorporate chest X-ray analysis alongside clinical notes, the architecture would need to evolve into a **multimodal fusion system**. Conceptually, this requires three components: an image encoder, a text encoder, and a fusion mechanism.

The chest X-ray images would be processed by a pre-trained convolutional neural network, a ResNet or DenseNet variant pre-trained on large radiology datasets such as CheXNet or ChestX-ray14. This produces a fixed-length embedding vector representing the visual features of the image: infiltrates, consolidation, pleural effusion, cardiomegaly.

The clinical notes would continue to be processed by ClinicalBERT, producing the [CLS] embedding as before. Both embeddings, image and text, would then be passed to a **fusion layer**: either a simple concatenation followed by a classification head, a cross-attention mechanism where image features attend to text features and vice versa, or a dedicated multimodal transformer such as CLIP-style contrastive learning adapted to the clinical domain. The fused representation would then drive the mortality prediction head.

This architecture mirrors the way a consulting physician integrates information: they read the notes, review the imaging, and form a unified clinical impression. The technical challenge lies in aligning the two modalities in a shared embedding space, a problem at the frontier of clinical AI research, but one for which validated approaches now exist in the literature.

End of Report — Part B

Appendix: Unified Model Comparison Table

Model	Accuracy	Precision	Recall ★	F1-Score	Train Time
DNN (Adam)	0.8618	0.3317	0.5681	0.4189	—
DNN (SGD)	0.8301	0.2921	0.6583	0.4046	—
LSTM	0.7721	0.2445	0.7652	0.3706	59.8s
GRU ★	0.7735	0.2482	0.7799	0.3765	74.7s
Bi-LSTM	0.7820	0.2523	0.7568	0.3784	97.2s
ClinicalBERT (Frozen)	0.9017	0.3030	0.0952	0.1449	279.2s
ClinicalBERT (Full)	0.7067	0.1841	0.6857	0.2903	645.8s

⭐ *GRU recommended for real-time ICU deployment based on highest deployable Recall (0.7799) and efficient inference.*